Neural Layer Matrices and Activations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Consider a dataset with n data points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, where each $\mathbf{x}_i \in \mathbb{R}^p$. The sample covariance matrix is given by $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$, where $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i$ is the sample mean. Suppose PCA is performed on this dataset, resulting in principal components defined by the eigenvectors of $\mathbf{S}$ and corresponding variances given by the eigenvalues. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ be the orthonormal eigenvectors of $\mathbf{S}$ with associated eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$. The first principal component captures the direction of maximum variance, λ_1, and the second principal component captures the next largest variance, λ_2, orthogonal to the first. If we decide to reduce the dimensionality from p to k by keeping only the first k principal components, the proportion of the total variance retained is given by $\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{p}\lambda_i}$. Suppose for a particular dataset, the eigenvalues of the covariance matrix are $\{10.5, 5.2, 2.1, 0.8, 0.4\}$. If we aim to retain at least 90% of the total variance, what is the minimum number of principal components required?

The total variance in the dataset is the sum of all eigenvalues: $\sum_{i=1}^{p}\lambda_i = 10.5 + 5.2 + 2.1 + 0.8 + 0.4 = 19.0$. We need to find the minimum k such that $\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{p}\lambda_i} \geq 0.90$. Let's check for different values of k: For $k = 1$: The variance retained is $\frac{10.5}{19.0} \approx 0.553$, which is less than 0.90. For $k = 2$: The variance retained is $\frac{10.5 + 5.2}{19.0} = \frac{15.7}{19.0} \approx 0.826$, which is less than 0.90. For $k = 3$: The variance.

2 **[Foundational]** • **Target: Affine Transformations and Matrix Dimensions**

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1?

Total variance is $\sum\lambda_i = 62 + 24 + 14 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum\lambda_i} = \frac{62}{100} = 62.0\%$. Final Answer: 62.0%

3 **[Foundational]** • **Target: Neural Parameter Counting**

A dataset consists of three independent observations X_1, X_2, X_3 drawn from a Poisson distribution with parameter λ. The observed values are $x_1 = 3, x_2 = 5, x_3 = 4$. We want to find the Maximum Likelihood Estimate (MLE) for λ. The log-likelihood function for a Poisson distribution is given by $\ell(\lambda) = \sum_{i=1}^{n}(x_i\log(\lambda) - \log(x_i!) - \lambda)$. Part 1: Derive the expression for the derivative of the log-likelihood function with respect to λ. Part 2: Set the derivative to zero to find the MLE of λ.

The log-likelihood function for $n = 3$ observations is $\ell(\lambda) = (x_1\log(\lambda) - \log(x_1!) - \lambda) + (x_2\log(\lambda) - \log(x_2!) - \lambda) + (x_3\log(\lambda) - \log(x_3!) - \lambda)$. Part 1: The derivative of the log-likelihood function with respect to λ is: $\frac{d\ell}{d\lambda} = \sum_{i=1}^{3}\left(\frac{x_i}{\lambda} - 1\right) = \left(\frac{x_1}{\lambda} - 1\right) + \left(\frac{x_2}{\lambda} - 1\right) + \left(\frac{x_3}{\lambda} - 1\right)$. Part 2: To find the MLE.

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4 [Foundational] • Target: Rectified Activations and Dying Neurons

A dataset consists of $n = 100$ independent and identically distributed (i.i.d.) observations $x_1, x_2, \ldots, x_{100}$. These observations are drawn from a Bernoulli distribution with parameter p. The probability mass function (PMF) of a Bernoulli distribution is given by $P(X = x; p) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$. The likelihood function $L(p)$ for this dataset is the product of the PMFs of the individual observations. If we observe $k = 60$ successes (i.e., $x_i = 1$) and $n - k = 40$ failures (i.e., $x_i = 0$), what is the likelihood function $L(p)$ for p?

The likelihood function $L(p)$ is the product of the probabilities of observing each data point given the parameter p. For i.i.d. Bernoulli trials, $L(p) = \prod_{i=1}^{n} P(X = x_i; p)$. Given $n = 100$ observations, with $k = 60$ successes (where $x_i = 1$) and $n - k = 40$ failures (where $x_i = 0$), the likelihood function can be written as: $L(p) = \left(\prod_{i:x_i=1} p^1(1-p)^{1-1}\right) \times \left(\prod_{i:x_i=0} p^0(1-p)^{1-0}\right)$.

5 [Foundational] • Target: Saturating Activations and Softmax Probabilities

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by: $\Sigma = (10, 5; 5, 20)$ What is the variance of the first feature, X_1?

The covariance matrix Σ for a dataset with two features X_1 and X_2 is structured as follows: $\Sigma = (Cov(X_1, X_1), Cov(X_1, X_2); Cov(X_2, X_1), Cov(X_2, X_2))$ By definition, the variance of a random variable is the covariance of the variable with itself. Therefore, $Var(X_1) = Cov(X_1, X_1)$ and $Var(X_2) = Cov(X_2, X_2)$. In the given covariance matrix: $\Sigma = (10, 5; 5, 20)$ The element in the first row and first column represents $Cov(X_1, X_1)$, which is the variance.

6 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Accuracy is the ratio of correct predictions to total samples: Accuracy $= \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 40}{130} = \frac{110}{130} = 84.6\%$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

7 [Foundational] • Target: Affine Transformations and Matrix Dimensions

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{50}$ to $\mathbf{y} \in \mathbb{R}^{25}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

The weight matrix $\mathbf{W}$ has shape 25×50, giving 1250 weights. The bias vector $\mathbf{b}$ has 25 parameters. Total = $1250 + 25 = 1275$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Neural Parameter Counting**

Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario?

The problem statement explicitly defines that the number of defective items, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The rate parameter λ is a direct characteristic of the Poisson distribution that determines its mean and variance. Therefore, the value of λ is directly given as 3.

9 **[Intermediate]** • **Target: Rectified Activations and Dying Neurons**

Consider two events, A and B, with the following probabilities: $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. What is the conditional probability of A given B, denoted as $P(A|B)$?

The formula for conditional probability is $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Given $P(A \cap B) = 0.2$ and $P(B) = 0.5$, we can substitute these values into the formula: $P(A|B) = \frac{0.2}{0.5} = 0.4$. Therefore, the conditional probability of A given B is 0.4.

10 **[Intermediate]** • **Target: Saturating Activations and Softmax Probabilities**

A centered feature matrix $\mathbf{X}$ has shape $N \times 22$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

Since $\mathbf{X}$ has 22 columns, $\mathbf{X}^T$ is $22 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 22×22. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

11 **[Intermediate]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Consider the scalar loss function $L(w) = 6w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 10$.

Differentiate with respect to w: $\frac{dL}{dw} = 2(6)w - 8 = 12w - 8$. At $w = 10$: $ablaL = 12(10) - 8 = 120 - 8 = 112$. Final Answer: $ablaL(10) = 112$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

12 **[Intermediate]** • **Target: Affine Transformations and Matrix Dimensions**

Given $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{36}{12} = 3$. Final Answer: $\hat{\mu}_{\text{MLE}} = 3$

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13 **[Intermediate]** • **Target: Neural Parameter Counting**

A medical diagnostic test for a rare disease is designed to be highly sensitive and specific. The disease affects 1 in every 1000 individuals in the population. The test has a sensitivity of 99% (meaning it correctly identifies 99% of individuals who have the disease) and a specificity of 95% (meaning it correctly identifies 95% of individuals who do not have the disease). If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease?

Let D be the event that an individual has the disease, and P be the event that the individual tests positive. We are given the following probabilities: - Prior probability of having the disease: $P(D) = \frac{1}{1000} = 0.001$ - Prior probability of not having the disease: $P(D^c) = 1 - P(D) = 1 - 0.001 = 0.999$ - Sensitivity (true positive rate): $P(P|D) = 0.99$ - Specificity (true negative rate): $P(P^c|D^c) = 0.95$ From specificity, we can derive the false positive rate.

14 **[Intermediate]** • **Target: Rectified Activations and Dying Neurons**

Consider a dataset of n data points, each residing in $\mathbb{R}^D$. We are performing Principal Component Analysis (PCA) to reduce the dimensionality. The first three principal components capture 70% of the total variance, the first six capture 85%, and the first ten capture 92%. You are tasked with selecting a subset of principal components for a downstream machine learning model. If the model requires that at least 90% of the original variance be retained, and you want to use the minimum possible number of principal components, what is the dimensionality of the transformed data?

The problem requires us to find the minimum number of principal components such that the cumulative explained variance is at least 90%. The provided information states: - The first 3 principal components capture 70% of the variance. - The first 6 principal components capture 85% of the variance. - The first 10 principal components capture 92% of the variance. We need to find the smallest k such that the cumulative variance captured by the first k principal components is $\geq 90\%$. Checking the giv.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **Target: Saturating Activations and Softmax Probabilities** • **Cross-Lesson: 3.01 Gradient Descent and Optimization**

A medical test for a rare disease has a 99\% accuracy rate in detecting the disease when it is present (True Positive Rate) and a 95\% accuracy rate in identifying individuals without the disease (True Negative Rate). If the prevalence of the disease in the population is 0.5\%, what is the probability that a person who tests positive actually has the disease?

Let D be the event that a person has the disease, and P be the event that the person tests positive. We are given: P(P|D) = 0.99 (True Positive Rate) P(not P|not D) = 0.95 (True Negative Rate), which means P(P|not D) = 1 - 0.95 = 0.05 (False Positive Rate) P(D) = 0.005 (Prevalence of the disease) From P(D), we know P(not D) = 1 - P(D) = 1 - 0.005 = 0.995. We want to find P(D|P), the probability that a person has the disease given a positive test result. Using Bayes' Theorem: P(D|P) = [P(P|D) * P(D)] / P(P) To find.

16 **[Challenging]** • **Target: Neural Layer Matrices & Activations Practice Exercises** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

Consider a binary classification problem where the decision boundary is defined by the logistic function $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$, which represents the probability of a data point $\mathbf{x} = (x_1, x_2)$ belonging to class 1. The estimated model parameters are $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$. We are interested in the geometric shape of the classification boundary. The classification boundary occurs where $p(x) = 0.5$. Find the equation of this boundary in the x_1x_2-plane.

The logistic regression model gives the probability of belonging to class 1 as $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$. The classification boundary is the set of points where $p(x) = 0.5$. Setting $p(x) = 0.5$, we have: $0.5 = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$ This implies: $1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} = 2$ Therefore: $e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} = 1$ Taking the natural logarithm of both sides.

17 **[Challenging]** • **Target: Affine Transformations and Matrix Dimensions** • **Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.

The given loss function is a quadratic function $L(w) = w^2 - 4w + 3$. This represents a parabola opening upwards. The minimum value of a quadratic function of the form $aw^2 + bw + c$ occurs at $w = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -4$. Therefore, the minimum occurs at $w = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Alternatively, we can find the minimum by taking the derivative and setting it to zero: $L'(w) = 2w - 4$. Setting $L'(w) = 0$ gives $2w - 4 = 0$, which implies $2w = 4$, so $w = 2$.

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18 **[Challenging]** • **Target: Neural Parameter Counting** • **Cross-Lesson: 6.01 Linear Regression and OLS**

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function?

To find the minimum of the quadratic loss function $L(w) = 3w^2 - 12w + 10$, we can find the derivative with respect to w and set it to zero. The derivative of $L(w)$ with respect to w is: $\frac{dL}{dw} = 6w - 12$. Setting the derivative to zero to find the critical point: $6w - 12 = 0$ $6w = 12$ $w = \frac{12}{6}$ $w = 2$. The second derivative is $\frac{d^2L}{dw^2} = 6$, which is positive, confirming that $w = 2$ corresponds to a minimum.

19 **[Challenging]** • **Target: Rectified Activations and Dying Neurons** • **Cross-Lesson: 7.01 Logistic Regression and Classification**

Consider a simple quadratic loss function $L(w) = (w^2 - 4w + 5)$. We want to find the minimum of this loss function using gradient descent. The gradient of the loss function with respect to w is given by $\frac{dL}{dw} = 2w - 4$. If we start at an initial weight $w_0 = 5$ and use a learning rate $\alpha = 0.5$, what will be the value of the weight w_1 after the first update step, i.e., $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$?

The loss function is $L(w) = w^2 - 4w + 5$. The gradient is $\frac{dL}{dw} = 2w - 4$. We are given an initial weight $w_0 = 5$ and a learning rate $\alpha = 0.5$. The gradient at w_0 is $\frac{dL}{dw}(w_0) = 2(5) - 4 = 10 - 4 = 6$. The update rule for gradient descent is $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$. Substituting the values, we get $w_1 = 5 - 0.5 \times 6 = 5 - 3 = 2$. Final Answer: 4

20 **[Challenging]** • **Target: Saturating Activations and Softmax Probabilities** • **Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -4$ and $z_2 = 6$.

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-4) = 0$ and $f(6) = 6$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{10}$ to $\mathbf{y} \in \mathbb{R}^{5}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. **55** ✓

b. 50

c. 15

d. 65

The weight matrix $\mathbf{W}$ has shape 5×10, giving 50 weights. The bias vector $\mathbf{b}$ has 5 parameters. Total = $50 + 5 = 55$.

22 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{20}$ to $\mathbf{y} \in \mathbb{R}^{10}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. 30

b. 200

c. **210** ✓

d. 230

The weight matrix $\mathbf{W}$ has shape 10×20, giving 200 weights. The bias vector $\mathbf{b}$ has 10 parameters. Total = $200 + 10 = 210$.

23 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{30}$ to $\mathbf{y} \in \mathbb{R}^{15}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. 45

b. 450

c. **465** ✓

d. 495

The weight matrix $\mathbf{W}$ has shape 15×30, giving 450 weights. The bias vector $\mathbf{b}$ has 15 parameters. Total = $450 + 15 = 465$.

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24 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{40}$ to $\mathbf{y} \in \mathbb{R}^{20}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. 60 **b.** 800

c. **820** ✓ **d.** 860

The weight matrix $\mathbf{W}$ has shape 20×40, giving 800 weights. The bias vector $\mathbf{b}$ has 20 parameters. Total = $800 + 20 = 820$.

25 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{50}$ to $\mathbf{y} \in \mathbb{R}^{25}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. **1275** ✓ **b.** 1250

c. 75 **d.** 1325

The weight matrix $\mathbf{W}$ has shape 25×50, giving 1250 weights. The bias vector $\mathbf{b}$ has 25 parameters. Total = $1250 + 25 = 1275$.

26 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{60}$ to $\mathbf{y} \in \mathbb{R}^{30}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. **1830** ✓ **b.** 1800

c. 90 **d.** 1890

The weight matrix $\mathbf{W}$ has shape 30×60, giving 1800 weights. The bias vector $\mathbf{b}$ has 30 parameters. Total = $1800 + 30 = 1830$.

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27 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Evaluate the output of the standard ReLU activation function \(f(z) = \max(0, z) \) for inputs \(z_1 = -2 \) and \(z_2 = 3 \).

a. \(f(z_1) = 2, \; f(z_2) = 3 \)

b. \(f(z_1) = -2, \; f(z_2) = 3 \)

c. \(f(z_1) = 0, \; f(z_2) = 0 \)

d. **\(f(z_1) = 0, \; f(z_2) = 3 \)** ✓

By definition of the Rectified Linear Unit: \(\text{ReLU}(z) = z \) if \(z > 0 \), and \(0 \) if \(z \le 0 \). Thus \(f(-2) = 0 \) and \(f(3) = 3 \).

28 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Evaluate the output of the standard ReLU activation function \(f(z) = \max(0, z) \) for inputs \(z_1 = -4 \) and \(z_2 = 6 \).

a. \(f(z_1) = 4, \; f(z_2) = 6 \)

b. \(f(z_1) = -4, \; f(z_2) = 6 \)

c. \(f(z_1) = 0, \; f(z_2) = 0 \)

d. **\(f(z_1) = 0, \; f(z_2) = 6 \)** ✓

By definition of the Rectified Linear Unit: \(\text{ReLU}(z) = z \) if \(z > 0 \), and \(0 \) if \(z \le 0 \). Thus \(f(-4) = 0 \) and \(f(6) = 6 \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Affine Transformations and Matrix Dimensions

A fully-connected neural layer maps input $\mathbf{x} \in \mathbf{R}^{128}$ to output $\mathbf{y} \in \mathbf{R}^{64}$ via $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. What is the shape of the weight matrix $\mathbf{W}$?

a. 64×128 ✓

b. 128×64

c. 128×128

d. 64×64

To multiply a (128×1) input and yield a (64×1) output, $\mathbf{W}$ must have shape 64×128.

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30 [Intermediate] • Target: Affine Transformations and Matrix Dimensions

For a mini-batch of $B = 32$ samples with input dimension $D = 128$, what is the shape of the activation batch $\mathbf{Y} = \mathbf{X}\mathbf{W}^T$ when $\mathbf{W}$ is 64×128?

a. 32×64 ✓

b. 64×32

c. 32×128

d. 128×64

$\mathbf{X}$ is (32×128), $\mathbf{W}^T$ is (128×64). Product shape: 32×64.

31 [Intermediate] • Target: Affine Transformations and Matrix Dimensions

If a two-layer linear network computes $\mathbf{y} = \mathbf{W}_2(\mathbf{W}_1\mathbf{x})$ without non-linear activations, why is its expressive capacity limited to a single linear transformation?

a. **The product of two matrices $\mathbf{W}_3 = \mathbf{W}_2\mathbf{W}_1$ is itself a single matrix** ✓

b. The gradient of two matrices is always undefined

c. Matrix multiplication is non-associative

d. The bias vector becomes zero

Linear mappings are closed under composition: $\mathbf{W}_2(\mathbf{W}_1\mathbf{x}) = (\mathbf{W}_2\mathbf{W}_1)\mathbf{x} = \mathbf{W}_{eq}\mathbf{x}$.

32 [Intermediate] • Target: Neural Parameter Counting

Calculate the total number of trainable parameters (weights + biases) in a dense layer with 256 inputs and 128 outputs.

a. 32928 ✓

b. 32768

c. 33024

d. 32896

Weights: $256 \times 128 = 32768$. Biases: 128. Total $= 32768 + 128 = 32928$.

33 [Intermediate] • Target: Neural Parameter Counting

A two-layer perceptron has architecture $10 \to 20 \to 1$. How many total parameters are in the network (including all biases)?

a. 241 ✓

b. 220

c. 200

d. 231

Layer 1: $10 \times 20 + 20 = 220$. Layer 2: $20 \times 1 + 1 = 21$. Total $= 220 + 21 = 241$.

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34 [Intermediate] • Target: Neural Parameter Counting

A convolutional layer has 32 filters of size 3×3 across 3 input channels with a bias per filter. How many learnable parameters does it contain?

a. 896 ✓

b. 864

c. 288

d. 928

Weights: $32 \times (3 \times 3 \times 3) = 32 \times 27 = 864$. Biases: 32. Total $= 864 + 32 = 896$.

35 [Intermediate] • Target: Rectified Activations and Dying Neurons

Evaluate the standard ReLU activation function $\mathrm{ReLU}(z) = \max(0, z)$ and its derivative for $z = -3.5$.

a. $\mathrm{ReLU}(-3.5) = 0, \quad \mathrm{ReLU}'(-3.5) = 0$ ✓

b. $\mathrm{ReLU}(-3.5) = -3.5, \quad \mathrm{ReLU}'(-3.5) = 1$

c. $\mathrm{ReLU}(-3.5) = 0, \quad \mathrm{ReLU}'(-3.5) = 1$

d. $\mathrm{ReLU}(-3.5) = 3.5, \quad \mathrm{ReLU}'(-3.5) = -1$

For $z < 0$, $\mathrm{ReLU}(z) = 0$ and its derivative is 0.

36 [Intermediate] • Target: Rectified Activations and Dying Neurons

How does the Leaky ReLU activation $f(z) = \max(0.01z, z)$ mitigate the 'dying ReLU' problem during backpropagation?

a. **It provides a non-zero gradient (0.01) for negative inputs, preventing weights from freezing** ✓

b. It normalizes inputs to have zero mean and unit variance

c. It bounds outputs between -1 and 1

d. It eliminates the need for bias vectors

For $z < 0$, $f'(z) = 0.01 > 0$, allowing error gradients to flow back through inactive neurons.

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37 [Intermediate] • Target: Rectified Activations and Dying Neurons

Find the output of a Leaky ReLU activation with coefficient $\alpha = 0.1$ when input is $z = -8$.

a. -0.8 ✓

b. 0

c. -8.0

d. 0.8

$f(-8) = 0.1(-8) = -0.8$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 6.01 Linear Regression and OLS

The standard logistic sigmoid is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.8$, compute the derivative $\sigma'(z_0)$.

a. 0.16 ✓

b. 0.80

c. 0.64

d. 0.20

$\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Here $\sigma'(z_0) = 0.8(1 - 0.8) = 0.8(0.2) = 0.16$.

39 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 7.01 Logistic Regression and Classification

Why does the Hyperbolic Tangent activation $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ generally train faster than standard sigmoid in hidden layers?

a. **It is zero-centered with outputs in $(-1, 1)$, preventing systematic gradient bias** ✓

b. Its derivative is identically 1 everywhere

c. It does not require exponent calculation

d. It eliminates all negative activations

Being zero-centered means average outputs are near 0, preventing zig-zag gradient updates across positive coordinates.

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40 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Apply the Softmax function $\text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum e^{z_j}}$ to the logit vector $\mathbf{z} = [0, \ln 2, \ln 5]^T$. Find the probability of class 3.

a. $\frac{5}{8} = 0.625$ ✓

b. $\frac{1}{8} = 0.125$

c. $\frac{2}{8} = 0.250$

d. $\frac{5}{7} \approx 0.714$

$e^0 = 1, e^{\ln 2} = 2, e^{\ln 5} = 5$. Sum $= 1 + 2 + 5 = 8$. $P(\text{class } 3) = 5/8 = 0.625$.

41 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.20$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.1600 ✓

b. 0.4000

c. 0.8000

d. 0.0400

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.20$:

$\sigma'(z_0) = (0.20)(1 - 0.20) = (0.20)(0.80) = 0.1600$.

42 [Challenging] • Target: Affine Transformations and Matrix Dimensions • Cross-Lesson: 3.01 Gradient Descent and Optimization

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

a. 2.4000

b. -0.2400 ✓

c. -0.2000

d. 1.4400

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 1.20$:

$\sigma'(z_0) = (1.20)(1 - 1.20) = (1.20)(-0.20) = -0.2400$.

43 [Challenging] • Target: Neural Parameter Counting • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.40$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.2400 ✓

b. 0.8000

c. 0.6000

d. 0.1600

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.40$:

$\sigma'(z_0) = (0.40)(1 - 0.40) = (0.40)(0.60) = 0.2400$.

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44 [Challenging] • Target: Rectified Activations and Dying Neurons • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.60$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.2400 ✓

b. 1.2000

c. 0.4000

d. 0.3600

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.60$:

$\sigma'(z_0) = (0.60)(1 - 0.60) = (0.60)(0.40) = 0.2400$.

45 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 6.01 Linear Regression and OLS

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.80$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.1600 ✓

b. 1.6000

c. 0.2000

d. 0.6400

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.80$:

$\sigma'(z_0) = (0.80)(1 - 0.80) = (0.80)(0.20) = 0.1600$.

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